

# MODEL QUESTION PAPER

**TED (15)6072**

## **SIXTH SEMESTER DIPLOMA EXAMINATION IN CHEMICAL ENGINEERING**

### **MASS TRANSFER OPERATIONS-II**

(Graph sheet should be issued)

Time: 3hrs

Maximum marks: 100

#### **PART –A**

Maximum marks: 10

- I Answer the following questions in one or two sentences. Each question carries 2 marks
- |   |                                                              |   |
|---|--------------------------------------------------------------|---|
| 1 | Define leaching                                              | 2 |
| 2 | Differentiate between leaching and liquid-liquid extraction. | 2 |
| 3 | Define super saturation.                                     | 2 |
| 4 | State the functions of agitators in crystallizer.            | 2 |
| 5 | Define flooding in sieve plate column.                       | 2 |

#### **PART -B**

- II Answer any five of the following questions. Each question carries 6 marks.
- |   |                                                                            |   |
|---|----------------------------------------------------------------------------|---|
| 1 | List the applications of leaching.                                         | 6 |
| 2 | Illustrate the working of Agitated vessel for leaching operations.         | 6 |
| 3 | Explain the working of packed column for liquid – liquid extraction.       | 6 |
| 4 | Suggest the super saturation methods in crystallization.                   | 6 |
| 5 | Illustrate the working of agitated tank crystallizer.                      | 6 |
| 6 | Summarize the mechanism of fractionation.                                  | 6 |
| 7 | List the feed conditions and corresponding values of 'q' in fractionation. | 6 |

PART -C  
Maximum Marks :60

Answer one full question from each unit. Each full question carries 15 marks

Module I

- |     |   |                                              |   |
|-----|---|----------------------------------------------|---|
| III | a | Explain the working of Bonnoto extractor.    | 8 |
|     | b | Summarise the continuous decantation system. | 7 |

OR

- |    |   |                                                                       |   |
|----|---|-----------------------------------------------------------------------|---|
| IV | a | Summarise the working of percolation tank.                            | 7 |
|    | b | Explain the constructional details and working of Bollmann extractor. | 8 |

Module II

- |   |   |                                                                    |   |
|---|---|--------------------------------------------------------------------|---|
| V | a | List the applications of liquid –liquid extraction.                | 7 |
|   | b | Explain the working of spray column for liquid –liquid extraction. | 8 |

OR

- |    |   |                                                                                  |   |
|----|---|----------------------------------------------------------------------------------|---|
| VI | a | Illustrate the working of Rotating disc contactor for liquid –liquid extraction. | 8 |
|    | b | Describe the working of batch mixer settler for liquid –liquid extraction.       | 7 |

Module III

- |     |   |                                                                                                                                                                                                                                                                                                                                                      |   |
|-----|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| VII | a | A hot solution containing 5000Kg of $MgSO_4$ and water at 420K and with a concentration of 50% by weight is cooled to 300K and $Mg SO_4 \cdot 10H_2O$ crystals is removed. The solubility at 300K is 30 Kg $MgSO_4$ per 100Kg water. Calculate the yield of crystals. Assume that no water is vapourised.( Atomic weight of Mg=24 , S=32 ,O=16 ,H=1) | 8 |
|     | b | Illustrate the working of Krystal crystallizer.                                                                                                                                                                                                                                                                                                      | 7 |

OR

- |      |   |                                                                                                                                                                                                                                                                                                                                                                     |   |
|------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| VIII | a | A hot solution containing 1000kg of $Na_2CO_3$ and water with a concentration of 30% by weight. $Na_2CO_3$ is cooled to 270 K and crystals of $Na_2CO_3 \cdot 10H_2O$ are precipitated. At 270K the solubility is 17Kg/100Kg water. Compute the yield of $Na_2CO_3$ crystals obtained. If 10% of the original water in the system evaporates on cooling. Data given | 8 |
|      |   | Molecular weight of $Na_2CO_3$ =106                                                                                                                                                                                                                                                                                                                                 |   |
|      |   | Molecular weight of, $Na_2CO_3 \cdot 10H_2O$ =286                                                                                                                                                                                                                                                                                                                   |   |

- b Explain the mechanism of crystallization. 7

Module IV

- IX a Mixture containing 70 mole% toluene and 30 mole% benzene is distilled in a continuous column at ordinary pressure to give product containing 95 mole% benzene and the bottom containing 5 mole% benzene. Reflux ratio is 3. Feed is saturated liquid. Calculate the number of theoretical plate required and determine the feed plate. 9

|   |      |      |      |      |      |      |      |   |
|---|------|------|------|------|------|------|------|---|
| x | 0.06 | 0.16 | 0.26 | 0.38 | 0.51 | 0.66 | 0.82 | 1 |
| y | 0.13 | 0.31 | 0.45 | 0.6  | 0.72 | 0.83 | 0.92 | 1 |

- b Summarise the optimum reflux ratio. 6

OR

- X a The 1000 Kgmol of mixture containing 50 mole% N-heptane and 50 mole% N-octane is subjected to fractional distillation at atmospheric pressure to give product containing 90 mole% N-heptane and bottom contain 10 mole% N-heptane. If reflux ratio is 3. Feed is a saturated vapour. Compute the number of theoretical plate required and feed plate. 9

|   |      |      |      |      |      |      |
|---|------|------|------|------|------|------|
| X | 0.05 | 0.1  | 0.2  | 0.3  | 0.4  | 0.5  |
| Y | 0.27 | 0.42 | 0.57 | 0.66 | 0.72 | 0.78 |

- b Derive an equation for 'q' line for rectification. 6